Calibration Is Not Style Robustness: Primal–Dual Counterfactual RLCD with Conformal Selective Risk Control

Anonymous submission

Abstract

Automation pipelines gate on typed probabilities such as $P(\text{yes})$ from decision models trained with strictly proper scores (RLCD). We prove that proper-score training on a style-correlated mixture can yield a predictor that is calibrated on the mixture yet overconfident on markdown-decorated wrong answers, and that monotone recalibration cannot repair its cross-style ranking. On *Style-CalibBench*, a decision-native split of RM-Bench, our RLCD reimplementation on Qwen2.5-1.5B reaches mixture ECE 0.089 but markdown ECE 0.171 and concise-correct vs. markdown-incorrect accuracy (ECA) 0.536; temperature scaling leaves ECA unchanged. We propose **CC-RLCD**, a primal–dual objective that enforces within-label counterfactual style invariance and cross-style margins under a bounded Brier budget, paired with prompt-clustered conformal risk control. CC-RLCD raises ECA to 0.694 and lowers markdown ECE to 0.098 at unchanged accuracy, without penalizing markdown-correct answers, and adds +14.1 to +16.6 ECA points across four backbones (0.5B–8B).

1 Introduction

Agentic and automation pipelines increasingly replace free-form LLM judges with typed decision interfaces that emit calibrated probabilities—such as $P(\text{yes})$ for substantive validity—in one forward pass (TypeSafe AI 2026; Guo et al. 2026; Gao et al. 2026). Because downstream code branches directly on p_c to auto-accept, abstain, or escalate, probability calibration is an operational requirement (Guo et al. 2017; Epstein et al. 2026).

Calibration on an observational test mixture is nevertheless incomplete. Reward models routinely conflate verbosity, markdown, sycophancy, and jargon with substance (Liu et al. 2025; Bharadwaj et al. 2026; Kim, Oh, and Lee 2026). Training a decision readout with a strictly proper score (Brier 1950; Gneiting and Raftery 2007), as in RLCD, elicits $P(y_c{=}1 \mid x, r)$, which absorbs any stylistic cue $s(r)$ correlated with training labels; when style and correctness conflict, an aggregate-calibrated gate silently accepts formatted errors.

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Resolving this conflict is a multi-objective optimization (MOO) problem (Sener and Koltun 2018; Gu et al. 2026; Xu et al. 2026) in which the mechanics decide whether debiasing reaches p_c at all: monotone post-hoc scaling fixes mixture ECE without altering cross-style ordering; a style head behind a stop-gradient cannot change p_c; and unconstrained scalarization or Counterfactual Data Augmentation (CDA) buys cross-style accuracy with calibration and accuracy (Das and Dennis 1997).

We propose **CC-RLCD** (*Counterfactual Constrained RLCD*; Figure 1a), which solves a constrained primal–dual program (Chamon and Ribeiro 2020) that updates the shared trunk with counterfactual invariance and conflict-margin losses subject to an ε-bounded content Brier budget, and gates decisions with prompt-clustered Conformal Risk Control (CRC) (Angelopoulos et al. 2024). Our contributions are:

- **Theory & protocol.** Proposition 1 shows that the proper-score optimum under a style-correlated mixture can be calibrated on the mixture yet overconfident on markdown; *Style-CalibBench* reports RM-Bench Easy/Normal/Hard, ECA and its reverse, and mixture vs. slice ECE.
- **Method.** CC-RLCD couples counterfactual invariance and conflict margins under an adaptive dual Brier budget, with prompt-clustered CRC gating.
- **Evidence.** Across four backbones and ten baselines, CC-RLCD gives the best accuracy–ECA trade-off, with controls for metric overlap, compute, and style shifts.

2 Related Work

Style Bias in Preference Models. RM-Bench reveals sharp reward-model drops when concise correct answers face markdown errors (Liu et al. 2025), and counterfactual audits expose length, jargon, and sycophancy biases (Bharadwaj et al. 2026). Mitigations span causal augmentation (Kim, Oh, and Lee 2026), post-hoc bias regression (Huang et al. 2025), arena rescaling (Zhu et al. 2025), and RLHF calibration (Leng et al. 2025). Closest to us, Patterning (Wang, Donoway, and Murfet 2026) reweights preference pairs by susceptibility (+14.2 pp RM-Bench Hard) and SteerRM (Sun et al. 2026) suppresses style-related SAE features at inference (up to +13.2 pp); both debias scalar rankers on official RM-Bench. We train calibrated probabilities $p_c \in [0, 1]$ for

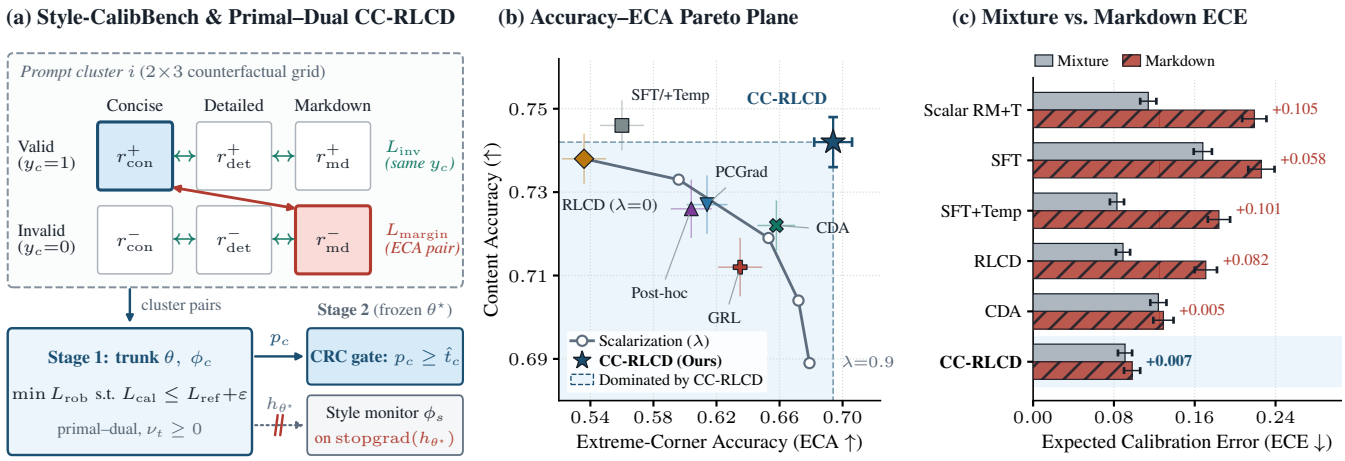


Figure 1: **Framework and 1.5B Main Results.** (a) *Style-CalibBench* crosses validity and style (2×3); Stage 1 of **CC-RLCD** updates (θ, ϕ_c) with invariance (L_{inv}) and conflict-margin (L_{margin}) losses under an ε-bounded Brier budget; Stage 2 sets the CRC threshold $\hat{t}_c$ (the detached style head is for monitoring only). (b) Accuracy–ECA plane (1 SE over 3 seeds; scalarization starts at RLCD, λ=0): all baseline point estimates lie in the region dominated by CC-RLCD, though the CDA cross overlaps (paired test in Section 5.1). (c) Mixture vs. markdown ECE ($+\Delta$ = MD-ECE − ECE).

selective gating and measure Hard on a prompt-disjoint RM-Bench split, so the numbers are not directly comparable.

Calibrated Decisions, MOO, and Risk Control. Proper scores elicit true conditionals (Gneiting and Raftery 2007) for System-One decision models (TypeSafe AI 2026; Guo et al. 2026; Gao et al. 2026; Deußer et al. 2026), which remain vulnerable to shift-induced overconfidence (Epstein et al. 2026). Multi-objective alignment uses gradient surgery (Yu et al. 2020), adversarial reversal (Ganin and Lempitsky 2015), value decorrelation (Xu et al. 2026), or Pareto filtering (Gu et al. 2026); we pair primal–dual constrained learning (Chamon and Ribeiro 2020) with CRC (Angelopoulos et al. 2024).

3 Theory and Style-CalibBench Protocol

3.1 Why Proper Scoring Fails Under Style Shift

Let prompt X, response R with content features $C = c(R)$ and style $S = s(R) \in \{\text{con}, \text{det}, \text{md}\}$, and validity $Y_c \in \{0, 1\}$. Brier or log loss drives $p_c(X, R)$ to $P_{\text{train}}(Y_c{=}1 \mid X, R)$, which, as S is a function of R, absorbs any training style–validity correlation instead of the style-invariant target $P(Y_c{=}1 \mid X, C)$.

Proposition 1 (Mixture-Calibration Cancellation)**.** *Fix $\delta \in (0, \frac{1}{4})$, let $S \in \{\text{con}, \text{md}\}$, and let $\pi = P(Y_c{=}1 \mid X, C) \sim \text{Unif}[\delta, 1-\delta]$. In training, let $P(S{=}\text{md} \mid Y_c{=}1, X, C) = \frac{\pi+\delta}{2\pi}$ and $P(S{=}\text{md} \mid Y_c{=}0, X, C) = \frac{1-\pi-\delta}{2(1-\pi)}$; at test, let S be uniform and independent of (Y_c, X, C). Then: (i) every strictly proper score over predictors of (X, C, S) is uniquely minimized by $p^\star = \pi + \delta$ on md and $\pi - \delta$ on con; (ii) on the test mixture, population ECE gives $\text{ECE}_{\text{mix}} = \frac{2\delta^2}{1-2\delta} < \delta = \text{ECE}_{\text{md}}$, and for any fixed binning B, $\text{ECE}^B_{\text{mix}} \le \frac{2\delta^2}{1-2\delta}$ while $\text{ECE}^B_{\text{md}} = \delta$; (iii) $\mathbb{E}[p^\star \mid Y_c{=}0, \text{md}] - \mathbb{E}[p^\star \mid Y_c{=}0, \text{con}] = 2\delta$; (iv) a same-prompt conflict pair $(r^+_{\text{con}}, r^-_{\text{md}})$ is misranked iff $\pi(x, c(r^+)) - \pi(x, c(r^-)) \le 2\delta$; (v) any strictly increasing $g : [0, 1] \to [0, 1]$ leaves ECA invariant.*

The proof is in Appendix A: for predicted values in $[2\delta, 1-2\delta]$ the md and con preimages $\pi = v \mp \delta$ have equal density and their gaps cancel, and binning cannot increase mixture ECE, so the gap also holds for our 15-bin ECE. Hence pooled reliability diagrams conceal markdown overconfidence, and temperature scaling repairs mixture ECE but leaves ECA unchanged.

3.2 Style-CalibBench and Metric Separation

Style-CalibBench adapts all 1,327 RM-Bench prompts (Liu et al. 2025) into 6 typed instances per prompt ($y_c \in \{1, 0\} \times s \in \{\text{con}, \text{det}, \text{md}\}$). On safety prompts, y_c=1 marks appropriate refusal or safe compliance. Prompt-level hashing (seed 20261109) yields 929 train, 133 calibration, and 265 test prompts (1,590 test instances) over Chat, Code, Math, Safety-refuse, and Safety-response. Dual-annotator review of 240 rewrites (κ=0.76) flagged 19 concise items dropping minor math qualifiers; we check stability with and without them.

We evaluate four axes: (1) **Calibration**: 15-bin mixture ECE and markdown slice ECE (MD-ECE). (2) **Cross-style ranking**: RM-Bench pair accuracies on p_c, where *Hard* averages $\text{con}^+ \succ \text{det}^-$, $\text{det}^+ \succ \text{md}^-$, $\text{con}^+ \succ \text{md}^-$, *Easy* averages the three style-reversed pairs, and *Normal* the same-style pairs; *Extreme-Corner Accuracy* $\text{ECA} = \frac{1}{N}\sum_i \mathbb{I}[p_c(x_i, r^+_{i,\text{con}}) > p_c(x_i, r^-_{i,\text{md}})]$; and its reverse *rECA* ($\text{md}^+ \succ \text{con}^-$), which exposes over-correction toward "markdown means wrong". (3) **Stability**: within-(x_i, y_c) 0.5-threshold *Flip* rate and *Confidence Inflation* $\Delta^y_{\text{md}} = \mathbb{E}[p_c \mid y, \text{md}] - \mathbb{E}[p_c \mid y, \text{con}]$ for $y \in \{0, 1\}$. (4) **Selective risk**: test error and coverage under the CRC threshold $\hat{t}_c$. CIs are

2,000-resample prompt-bootstrap 95% intervals. Since test prompts share the RM-Bench distribution with training and our margin loss targets the Hard pairs (Section 4), our Hard is in-distribution and not comparable to the public RM-Bench leaderboard.

4 Counterfactual Constrained RLCD

4.1 Detached Heads Cannot Change p_c

Let $h_\theta(x, r)$ denote the shared trunk representation, $u_c = \mathrm{softmax}(\phi_c(h_\theta)) \in \Delta_2$ the binary content distribution ($p_c = u_{c,1}$), and $u_s = \mathrm{softmax}(\phi_s(h_\theta)) \in \Delta_5$ a 1–5 match score for a declared target style s^* (gold level 5 if equal, 3 for det/md, 2 for con/det, 1 for con/md). A detached two-stage recipe minimizes the Brier risk $L_{\mathrm{cal}}(\theta, \phi_c) = \mathbb{E}[\|u_c - e_{y_c}\|_2^2]$ in Stage A and $L_s(\mathrm{stopgrad}(h_{\theta_A}), \phi_s)$ in Stage B. Since $\nabla_{\theta,\phi_c} L_s \equiv 0$, Stage B leaves (θ, ϕ_c), and hence p_c, unchanged; moving p_c requires counterfactual gradients in θ, traded off explicitly against calibration.

4.2 Stage 1: Primal–Dual Counterfactual Optimization

Within each training prompt cluster i, let $z_c = \log \frac{p_c}{1-p_c}$ denote the content log-odds and $\mathcal{P}_{\mathrm{adv}} = \{(\mathrm{con}, \mathrm{det}), (\mathrm{det}, \mathrm{md}), (\mathrm{con}, \mathrm{md})\}$ the style-conflict pairs (m=1.0). We define within-label invariance and cross-style margin losses:

$$L_{\mathrm{inv}} = \mathbb{E}_{i,y_c,s\neq s'}\big[(p_c(x_i, r_{i,s}^{y_c}) - p_c(x_i, r_{i,s'}^{y_c}))^2\big], \quad (1)$$

$$L_{\mathrm{margin}} = \mathbb{E}_{i,(a,b)}\big[\max\big(0,\, m - \Delta z_c(x_i; a, b)\big)\big],$$

where $\Delta z_c(x_i; a, b) = z_c(x_i, r_{i,a}^+) - z_c(x_i, r_{i,b}^-)$ for $(a, b) \in \mathcal{P}_{\mathrm{adv}}$ (the Hard pairs; see Section 5.1 for a held-out control). Setting $L_{\mathrm{rob}} = L_{\mathrm{inv}} + \beta L_{\mathrm{margin}}$ and anchoring to a Stage-0 RLCD Brier risk L_{ref}, we solve the constrained program:

$$\min_{\theta,\phi_c} L_{\mathrm{rob}}(\theta, \phi_c) \quad \text{s.t.} \quad L_{\mathrm{cal}}(\theta, \phi_c) \le L_{\mathrm{ref}} + \varepsilon, \quad (2)$$

via primal–dual updates on $\mathcal{L}(\theta, \phi_c, \nu) = L_{\mathrm{rob}}(\theta, \phi_c) + \nu(L_{\mathrm{cal}}(\theta, \phi_c) - L_{\mathrm{ref}} - \varepsilon)$ with projected dual ascent:

$$\nu_{t+1} = \big[\nu_t + \eta_\nu\big(\widehat{L}_{\mathrm{cal}}(\theta_t, \phi_{c,t}) - L_{\mathrm{ref}} - \varepsilon\big)\big]_+. \quad (3)$$

Unlike fixed scalarization $(1-\lambda)L_{\mathrm{cal}} + \lambda L_{\mathrm{rob}}$, ν_t grows only when debiasing threatens Brier honesty and relaxes inside $L_{\mathrm{ref}} + \varepsilon$, which can reach non-convex Pareto knees (Chamon and Ribeiro 2020).

4.3 Stage 2: Conformal Gate

On M=133 calibration clusters, let $\hat{R}_i(t) \in [0, 1]$ be the error rate among items of cluster i accepted at threshold t ($p_c \ge t$), with $\hat{R}_i(t) = 0$ if none is accepted, monotonized as $\sup_{t' \ge t} \hat{R}_i(t')$ so it is non-increasing with $\hat{R}_i(1) = 0$. Prompt-clustered CRC (Angelopoulos et al. 2024) sets:

$$\hat{t}_c = \inf\Big\{t \in [0, 1] : \frac{1}{M+1}\Big(\sum_{i=1}^{M} \hat{R}_i(t) + 1\Big) \le \alpha\Big\}, \quad (4)$$

Method (1.5B)	Acc↑	Hard↑	ECA↑	ECE↓	MD-ECE↓	Flip↓
Readout	0.547	0.407	0.358	0.241	0.327	0.519
Scalar RM+T†	0.618	0.459	0.412	0.114	0.219	0.463
SFT	**0.746**	0.611	0.560	0.168	0.226	0.337
SFT+Temp	**0.746**	0.611	0.560	**0.083**	0.184	0.337
Post-hoc Reg.	0.726	0.648	0.604	0.097	0.142	0.241
RLCD (Brier)	0.738	0.576	0.536	0.089	0.171	0.281
Detached 2-H*	0.738	0.576	0.536	0.089	0.171	0.281
PCGrad	0.727	0.637	0.614	0.111	0.156	0.214
GRL (Adv.)	0.712	0.662	0.635	0.119	0.138	0.189
CDA	0.722	0.684	0.658	0.124	0.129	0.171
CC-RLCD	0.742	**0.716**	**0.694**	0.091	**0.098**	**0.138**
w/o L_{inv}	0.736	0.695	0.671	0.098	0.115	0.169
w/o L_{margin}	0.740	0.668	0.639	0.090	0.109	0.147
Fixed λ=0.5	0.719	0.679	0.653	0.116	0.125	0.164

Table 1: Test means (3 seeds) on Qwen2.5-1.5B. †Off-the-shelf 8B reward model, not size-matched. *Stage-B style head on the RLCD checkpoint: style QWK 0.72, but over all 1,590 test items $\max|\Delta z_c| = 2.6 \times 10^{-6}$ and 0 ECA flips, as Section 4 predicts.

which guarantees $\mathbb{E}[\hat{R}_{M+1}(\hat{t}_c)] \le \alpha$ for a new exchangeable prompt cluster, so we report test risk as the same cluster-level quantity averaged over test clusters. The guarantee is marginal over clusters and does not cover the markdown slice. An optional style head ϕ_s on $\mathrm{stopgrad}(h_{\theta^\star})$ (5-level ordinal Brier loss) logs style match for monitoring only; it is not part of the gate.

5 Experiments

Setup and Baselines. We train Qwen2.5-1.5B-Instruct (LoRA r=16, lr 2×10^{-5}, length 1024, ε=0.005, β=0.5, η_ν=0.05, seeds {17, 23, 41}) and replicate on Qwen2.5-0.5B, Qwen2.5-7B, and Llama-3.1-8B-Instruct. All trained methods get the same 3-epoch budget on the 929 training clusters (CC-RLCD: 2 Stage-0 RLCD epochs + 1 primal–dual epoch, 1.08× RLCD wall-clock). Baselines: **Readout** (untuned model; p_c = softmax over its Yes/No logits); **Scalar RM+T** (off-the-shelf Skywork-Reward-Llama-3.1-8B, sigmoid temperature fit on calibration prompts; an 8B reference, not size-matched); **SFT** (Yes/No cross-entropy) and **SFT+Temp**; **Post-hoc Reg.** (length/markdown regression on SFT scores (Huang et al. 2025)); **RLCD (Brier)**; **Detached 2-Head**; and **PCGrad** (Yu et al. 2020), **GRL** (Ganin and Lempitsky 2015), and **CDA** (Kim, Oh, and Lee 2026) on the same clusters. As the RLCD recipe is unpublished (TypeSafe AI 2026), RLCD (Brier) is our reimplementation: p_c is read deterministically from the readout, so the negative-Brier reward is differentiable and RLCD reduces to minimizing L_{cal}; we do not claim to reproduce the proprietary recipe.

5.1 Main Comparison

Comparison with Baselines. Table 1 and Figure 1b–c corroborate Proposition 1. RLCD reaches mixture ECE 0.089 [0.067, 0.116] but markdown ECE 0.171 (concise 0.061) and ECA 0.536 [0.479, 0.592]; SFT+Temp reaches ECE

Method (1.5B)	Easy	Normal	Hard	ECA	rECA	Δ^1_{md}
SFT+Temp	0.866	0.781	0.611	0.560	0.903	+0.064
RLCD	0.874	0.776	0.576	0.536	0.912	+0.071
CDA	0.823	0.768	0.684	0.658	0.846	+0.012
CC-RLCD	0.838	0.792	0.716	0.694	0.857	−0.009
held-out corner	0.845	0.789	0.701	0.672	0.866	−0.006

Table 2: RM-Bench splits and reverse-direction checks (1.5B). *Held-out corner*: L_{margin} omits the ECA pair. $\Delta^1_{\text{md}} \ll 0$ would signal over-correction.

0.083 without moving ECA (0.560). Post-hoc regression lifts ECA to 0.604, and the shared-trunk baselines reach at most 0.658 (CDA) at lower accuracy (0.712–0.727) and higher ECE (0.111–0.124). **CC-RLCD** reaches ECA 0.694 [0.642, 0.746] and Hard 0.716, cuts markdown ECE to 0.098 and flips to 0.138, and keeps accuracy (0.742) and ECE (0.091). Its ECA gain is +0.158 over RLCD ($p<0.001$) but only +0.036 over CDA, the strongest baseline (paired prompt bootstrap [+0.004, +0.068], $p=0.031$); the clearer advantage over CDA is in calibration (MD-ECE −0.031, ECE −0.033; $p<0.01$) and accuracy (+0.020, $p=0.012$). Wrong-answer inflation Δ^0_{md} drops from +0.083 to +0.019. In ablations, L_{inv} drives flip stability, L_{margin} drives ECA (+5.5 pts), and the adaptive dual ($\nu^\star \approx 1.84$) beats fixed scalarization. Excluding the 19 drifted concise items moves every ECA by at most 0.012 and preserves the ranking.

Training–Metric Overlap and Over-Correction. Because L_{margin} trains on the Hard pairs, Table 2 adds three checks. (i) With the ECA pair held out of training, ECA still reaches 0.672 (0.639 without any margin), so most of the gain transfers from the adjacent pairs. (ii) Easy and rECA stay above Normal, and the markdown shift on correct answers is −0.009 [−0.024, +0.006]: CC-RLCD removes the markdown bonus rather than learning a markdown penalty. (iii) The Easy–Hard gap shrinks from 0.298 to 0.122, leaving residual style bias.

5.2 Conformal Risk, Style Shifts, and Scaling

Conformal Selective Prediction. Naive thresholding overshoots $\alpha=0.05$ (RLCD 0.061, SFT 0.078); prompt-clustered CRC (Eq. 4; Figure 2a–b) keeps cluster-averaged test risk ≤ 0.046 for all methods, as guaranteed (pooled item-level risk differs by ≤ 0.004). On markdown, where no guarantee applies, RLCD and SFT+Temp exceed 5% (0.086, 0.096); CC-RLCD reaches 0.049 [0.038, 0.061], at the target but with a CI crossing it. CC-RLCD also raises coverage from 0.351 to 0.436, whereas adding the style head as a gate lowers coverage (0.362) without reducing risk.

Slices, Style Shifts, and Scale. CC-RLCD improves 1.5B ECA in all five domains (+11.9 to +18.2 pts). For style shifts (Figure 2c), we apply three Bharadwaj et al. (2026) perturbations—*Length Swap*, *Sycophancy*, *Authoritative Jargon*—to the incorrect concise response of each test prompt with Qwen2.5-72B-Instruct and a fixed template, keeping rewrites an annotator confirms content-preserving

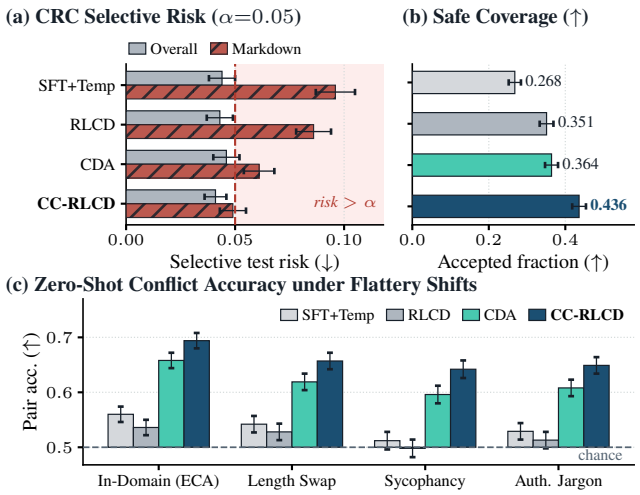


Figure 2: **Selective Risk and Style Shifts (1.5B).** (a–b) Prompt-clustered CRC ($\alpha=0.05$; 1 SE over 3 seeds); the markdown slice carries no guarantee. (c) Conflict-pair accuracy under *Flattery* perturbations; bars start at chance.

Backbone	Acc↑	ECA↑	ECE↓	MD-ECE↓
Qwen2.5-0.5B	.688→.701	.493→.634	.117→.114	.204→.126
Qwen2.5-7B	.784→.789	.581→.747	.074→.077	.152→.083
Llama-3.1-8B	.779→.783	.574→.739	.079→.081	.158→.086

Table 3: Scaling, RLCD→CC-RLCD (3 seeds). 7B CDA: Acc 0.769, ECA 0.708, ECE 0.106, MD-ECE 0.114.

(241/236/229 of 265 pairs). Length Swap overlaps the con–det training axis; the other two are unseen. RLCD stays near chance (0.498–0.528), whereas CC-RLCD reaches 0.642–0.657 (+14.4 and +13.6 pts on the unseen axes). Table 3 shows that 7B/8B scale does not remove style bias under RLCD (MD-ECE 0.152–0.158), while CC-RLCD adds +14.1 to +16.6 ECA points on every backbone; on 7B its gain over CDA is +0.039 (paired $p=0.024$).

6 Limitations and Conclusion

Our Hard is in-distribution and shares pair types with L_{margin}, which our controls mitigate but do not remove; ECA half-widths are ±5–6 pts; CRC bounds only cluster-averaged overall risk (a markdown guarantee needs Mondrian CRC); RLCD is our reimplementation; and Stage 1 needs open weights. Mixture calibration does not imply style robustness, and neither post-hoc scaling nor detached heads change cross-style rankings; primal–dual counterfactual training with CRC yields better-calibrated, more style-robust decision probabilities.

Ethical and AI Use Statement

We use RM-Bench under its license terms; style scores are monitoring diagnostics, not truth labels. LLMs assisted with prose editing and plotting.

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A Proof of Proposition 1

Write $\pi = \pi(X, C)$ and $a = \frac{\pi+\delta}{2\pi}$, $b = \frac{1-\pi-\delta}{2(1-\pi)}$; both lie in $[0, 1]$ because $\pi \in [\delta, 1-\delta]$.

(i) $P_{\text{train}}(S{=}\text{md} \mid X, C) = \pi a + (1-\pi)b = \frac{1}{2}$. By Bayes' rule, $P_{\text{train}}(Y_c{=}1 \mid X, C, \text{md}) = 2\pi a = \pi + \delta$ and $P_{\text{train}}(Y_c{=}1 \mid X, C, \text{con}) = 2\pi(1-a) = \pi - \delta$. A strictly proper score is uniquely minimized in expectation by the true conditional probability, hence by $p^\star$.

(ii) At test, S is independent of (Y_c, X, C), so $P_{\text{test}}(Y_c{=}1 \mid X, C, S) = \pi$. On the md slice $p^\star = \pi + \delta$ is a bijection of π, so $\mathbb{E}[Y_c \mid p^\star{=}v, \text{md}] = v - \delta$: the gap is exactly δ at every v, so $\text{ECE}_{\text{md}} = \delta$, and every bin's average gap is also δ, so $\text{ECE}^B_{\text{md}} = \delta$. On the mixture, $p^\star$ is uniform on $[2\delta, 1]$ (md) and on $[0, 1-2\delta]$ (con), each slice with weight $\frac{1}{2}$ and density $\frac{1}{1-2\delta}$. For $v \in [2\delta, 1-2\delta]$ both slices have equal density at v, so $\mathbb{E}[Y_c \mid p^\star{=}v] = \frac{1}{2}(v-\delta) + \frac{1}{2}(v+\delta) = v$. For $v < 2\delta$ only con contributes (gap δ), and for $v > 1-2\delta$ only md contributes (gap δ). Each end interval has mass $\frac{1}{2} \cdot \frac{2\delta}{1-2\delta}$, so $\text{ECE}_{\text{mix}} = \frac{2\delta^2}{1-2\delta}$, which is below δ iff $\delta < \frac{1}{4}$. For any bin b, $\left|\mathbb{E}[(Y_c - p^\star)\mathbb{I}_b]\right| \le \mathbb{E}\left[\left|\mathbb{E}[Y_c \mid p^\star] - p^\star\right| \mathbb{I}_b\right]$; summing over bins gives $\text{ECE}^B_{\text{mix}} \le \text{ECE}_{\text{mix}}$.

(iii) Conditional on $Y_c{=}0$ at test, S is still independent of (X, C), so π has the same conditional law in both slices and the difference is $(\pi + \delta) - (\pi - \delta) = 2\delta$ in expectation.

(iv) $p^\star(x, r^+_{\text{con}}) = \pi(x, c(r^+)) - \delta$ and $p^\star(x, r^-_{\text{md}}) = \pi(x, c(r^-)) + \delta$, so the strict ECA comparison fails iff $\pi(x, c(r^+)) - \pi(x, c(r^-)) \le 2\delta$.

(v) A strictly increasing g preserves every strict inequality $p_c(x, r^{+}_{\mathrm{con}}) > p_c(x, r^{-}_{\mathrm{md}})$ and its negation, so each ECA indicator is unchanged. □